Computing Publications with Major Personal Contributions

Oliver Gutsche
May 05, 2026

1. Building an AI-native Research Ecosystem for Experimental Particle Physics: A Community Vision
Aarrestad, Thea Klaeboe et al.
arXiv:2602.17582 | 2026-02-19

2. Interpretable Models for Workflow Differentiation in High-Performance Scientific Networks
Giannakou, Anna et al.
(2026) 349 | DOI:10.1109/ICNC68183.2026.11416957 | 2026

3. Evaluating Application Characteristics for GPU Portability Layer Selection
Atif, Mohammad et al.
arXiv:2601.17526 | 2026-01-24

4. Application of performance portability solutions for GPUs and many-core CPUs to track reconstruction kernels
Kwok, Ka Hei Martin et al.
EPJ Web Conf. 295 (2024) 11003 | arXiv:2401.14221 | DOI:10.1051/epjconf/202429511003 | 2024-01-25

5. The U.S. CMS HL-LHC R D Strategic Plan
Gutsche, Oliver et al.
EPJ Web Conf. 295 (2024) 04050 | arXiv:2312.00772 | DOI:10.1051/epjconf/202429504050 | 2023-12-01

6. A Ceph S3 Object Data Store for HEP
Smith, Nick et al.
EPJ Web Conf. 295 (2024) 01003 | arXiv:2311.16321 | DOI:10.1051/epjconf/202429501003 | 2023-11-27

7. Evaluating Portable Parallelization Strategies for Heterogeneous Architectures in High Energy Physics
Atif, Mohammad et al.
arXiv:2306.15869 | 2023-06-27

8. Detector R D needs for the next generation e^+e^- collider
Apresyan, A. et al.
arXiv:2306.13567 | 2023-06-23

9. Automated Network Services for Exascale Data Movement
Balcas, Justas et al.
EPJ Web Conf. 295 (2024) 01009 | DOI:10.1051/epjconf/202429501009 | 2024

10. IRIS-HEP Strategic Plan for the Next Phase of Software Upgrades for HL-LHC Physics
Bockelman, Brian et al.
arXiv:2302.01317 | 2023-02-02

11. The Future of High Energy Physics Software and Computing
Elvira, V. Daniel et al.
arXiv:2210.05822 | DOI:10.2172/1898754 | 2022-10-11

12. Snowmass Computational Frontier: Topical Group Report on Experimental Algorithm Parallelization

Cerati, G. et al.
arXiv:2209.07356 | 2022-09-15

13. Portability: A Necessary Approach for Future Scientific Software
Bhattacharya, Meghna et al.
arXiv:2203.09945 | 2022-03-15

14. Coffea: Columnar Object Framework For Effective Analysis
Smith, Nicholas et al.
EPJ Web Conf. 245 (2020) 06012 | arXiv:2008.12712 | DOI:10.1051/epjconf/202024506012 | 2020-08-28

15. Response to NITRD, NCO, NSF Request for Information on "Update to the 2016 National Artificial Intelligence Research and Development Strategic Plan"
Amundson, J. et al.
arXiv:1911.05796 | DOI:10.2172/1592156 | 2019-11-04

16. Striped Data Analysis Framework
Gutsche, Oliver et al.
EPJ Web Conf. 245 (2020) 06042 | DOI:10.1051/epjconf/202024506042 | 2020

17. CMS Computing Resources: Meeting the Demands of the High-Luminosity LHC Physics Program
Lange, David et al.
EPJ Web Conf. 214 (2019) 03055 | DOI:10.1051/epjconf/201921403055 | 2019-09-17

18. New Technologies for Discovery
Ahmed, Z. et al.
arXiv:1908.00194 | 2019-07-31

19. Using Big Data Technologies for HEP Analysis
Cremonesi, Matteo et al.
EPJ Web Conf. 214 (2019) 06030 | arXiv:1901.07143 | DOI:10.1051/epjconf/201921406030 | 2019-01-21

20. HEP Software Foundation Community White Paper Working Group – Data Organization, Management and Access (DOMA)
Berzano, Dario et al.
arXiv:1812.00761 | 2018-11-30

21. HEP Software Foundation Community White Paper Working Group – Visualization
Bellis, Matthew et al.
arXiv:1811.10309 | 2018-11-26

22. HPC resource integration into CMS Computing via HEPCloud
Hufnagel, Dirk et al.
EPJ Web Conf. 214 (2019) 03031 | DOI:10.1051/epjconf/201921403031 | 2018-10-16

23. HEP Software Foundation Community White Paper Working Group - Data Analysis and Interpretation
Bauerdick, Lothar et al.
arXiv:1804.03983 | 2018-04-09

24. A Roadmap for HEP Software and Computing R D for the 2020s
Albrecht, Johannes et al.
Comput.Softw.Big Sci. 3 (2019) 7 | arXiv:1712.06982 | DOI:10.1007/s41781-018-0018-8 | 2017-12-18

25. Experience in using commercial clouds in CMS
Bauerdick, L. et al.
J.Phys.Conf.Ser. 898 (2017) 052019 | DOI:10.1088/1742-6596/898/5/052019 | 2017-11-22

26. CMS Analysis and Data Reduction with Apache Spark
Gutsche, Oliver et al.
J.Phys.Conf.Ser. 1085 (2018) 042030 | arXiv:1711.00375 | DOI:10.1088/1742-6596/1085/4/042030 | 2017-10-31

27. HEPCloud, a New Paradigm for HEP Facilities: CMS Amazon Web Services Investigation
Holzman, Burt et al.
Comput.Softw.Big Sci. 1 (2017) 1 | arXiv:1710.00100 | DOI:10.1007/s41781-017-0001-9 | 2017-09-29

28. Striped Data Server for Scalable Parallel Data Analysis
Chang, Jin et al.
J.Phys.Conf.Ser. 1085 (2018) 042035 | DOI:10.1088/1742-6596/1085/4/042035 | 2017

29. Big Data in HEP: A comprehensive use case study
Gutsche, Oliver et al.
J.Phys.Conf.Ser. 898 (2017) 072012 | arXiv:1703.04171 | DOI:10.1088/1742-6596/898/7/072012 | 2017-03-12

30. Dark Matter and Super Symmetry: Exploring and Explaining the Universe with Simulations at the LHC
Gutsche, Oliver et al.
(2017) 4 | DOI:10.1109/WSC.2016.7822075 | 2017-01-19

31. ASCR/HEP Exascale Requirements Review Report
Habib, Salman et al.
arXiv:1603.09303 | 2016-03-30

32. Pooling the resources of the CMS Tier-1 sites
Apyan, A. et al.
J.Phys.Conf.Ser. 664 (2015) 042056 | DOI:10.1088/1742-6596/664/4/042056 | 2015-12-23

33. Using the GlideinWMS System as a Common Resource Provisioning Layer in CMS
Balcas, J. et al.
J.Phys.Conf.Ser. 664 (2015) 062031 | DOI:10.1088/1742-6596/664/6/062031 | 2015-12-23

34. Pushing HTCondor and glideinWMS to 200K+ Jobs in a Global Pool for CMS before Run 2
Balcas, J. et al.
J.Phys.Conf.Ser. 664 (2015) 062030 | DOI:10.1088/1742-6596/664/6/062030 | 2015-12-23

35. Diversity in Computing Technologies and Strategies for Dynamic Resource Allocation
Garzoglio, G. et al.
J.Phys.Conf.Ser. 664 (2015) 012001 | DOI:10.1088/1742-6596/664/1/012001 | 2015-12-23

36. Fermilab Computing at the Intensity Frontier
Group, Craig et al.
J.Phys.Conf.Ser. 664 (2015) 032012 | DOI:10.1088/1742-6596/664/3/032012 | 2015-12-23

37. High Energy Physics Forum for Computational Excellence: Working Group Reports (I. Applications Software II. Software Libraries and Tools III. Systems)
Habib, Salman et al.
arXiv:1510.08545 | 2015-10-28

38. CMS Computing Operations During Run 1
Adelman, J. et al.
J.Phys.Conf.Ser. 513 (2014) 032040 | DOI:10.1088/1742-6596/513/3/032040 | 2013-10-30

39. Evolution of the Pilot Infrastructure of CMS: Towards a Single GlideinWMS Pool
Belforte, S. et al.

J.Phys.Conf.Ser. 513 (2014) 032041 | DOI:10.1088/1742-6596/513/3/032041 | 2014

40. Deployment of a WLCG network monitoring infrastructure based on the perfSONAR-PS technology
Campana, S. et al.
J.Phys.Conf.Ser. 513 (2014) 062008 | DOI:10.1088/1742-6596/513/6/062008 | 2014

41. Opportunistic Resource Usage in CMS
Kreuzer, Peter et al.
J.Phys.Conf.Ser. 513 (2014) 062028 | DOI:10.1088/1742-6596/513/6/062028 | 2014

42. Towards a Centralized Grid Speedometer
Dzhunov, I. et al.
J.Phys.Conf.Ser. 513 (2014) 032028 | DOI:10.1088/1742-6596/513/3/032028 | 2014

43. CMS experience of running glideinWMS in High Availability mode
Sfiligoi, I. et al.
J.Phys.Conf.Ser. 513 (2014) 032086 | DOI:10.1088/1742-6596/513/3/032086 | 2014

44. A New Era for Central Processing and Production in CMS
Fajardo, E. et al.
J.Phys.Conf.Ser. 396 (2012) 042018 | DOI:10.1088/1742-6596/396/4/042018 | 2012-06-27

45. No File Left Behind - Monitoring Transfer Latencies in PhEDEx
Chwalek, T. et al.
J.Phys.Conf.Ser. 396 (2012) 032089 | DOI:10.1088/1742-6596/396/3/032089 | 2012

46. CMS Data Transfer Operations After the First Years of LHC Collisions
Kaselis, Rapolas et al.
J.Phys.Conf.Ser. 396 (2012) 042033 | DOI:10.1088/1742-6596/396/4/042033 | 2012

47. Monitoring Techniques and Alarm Procedures for CMS Services and Sites in WLCG
Molina-Perez, J. et al.
J.Phys.Conf.Ser. 396 (2012) 042041 | DOI:10.1088/1742-6596/396/4/042041 | 2012

48. CMS distributed computing workflow experience
Adelman-McCarthy, Jennifer et al.
J.Phys.Conf.Ser. 331 (2011) 072019 | DOI:10.1088/1742-6596/331/7/072019 | 2011

49. Experience Building and Operating the CMS Tier-1 Computing Centres
Albert, M. et al.
J.Phys.Conf.Ser. 219 (2010) 072035 | DOI:10.1088/1742-6596/219/7/072035 | 2010

50. Use of glide-ins in CMS for production and analysis
Bradley, D. et al.
J.Phys.Conf.Ser. 219 (2010) 072013 | DOI:10.1088/1742-6596/219/7/072013 | 2010

51. Validation of Software Releases for CMS
Gutsche, Oliver et al.
J.Phys.Conf.Ser. 219 (2010) 042040 | DOI:10.1088/1742-6596/219/4/042040 | 2009-03

52. Stand-alone Cosmic Muon Reconstruction Before Installation of the CMS Silicon Strip Tracker
Adam, W. et al.
JINST 4 (2009) P05004 | arXiv:0902.1860 | DOI:10.1088/1748-0221/4/05/P05004 | 2009-02

53. Large scale job management and experience in recent data challenges within the LHC CMS experiment
Evans, D. et al.

PoS ACAT08 (2008) 032 | DOI:10.22323/1.070.0032 | 2008

54. The CMS Remote Analysis Builder (CRAB)
Spiga, D. et al.
Lect.Notes Comput.Sci. 4873 (2007) 580 | DOI:10.1007/978-3-540-77220-0_52 | 2007

55. WLCG scale testing during CMS data challenges
Gutsche, Oliver et al.
J.Phys.Conf.Ser. 119 (2008) 062033 | DOI:10.1088/1742-6596/119/6/062033 | 2008

56. CRAB: The CMS distributed analysis tool development and design
Spiga, D. et al.
Nucl.Phys.B Proc.Suppl. 177-178 (2008) 267 | DOI:10.1016/j.nuclphysbps.2007.11.124 | 2008

57. Status and evolution of CRAB
Farina, Fabio et al.
PoS ACAT (2007) 020 | DOI:10.22323/1.050.0020 | 2007

58. A ROOT based client server event display for the ZEUS experiment
Kind, O. et al.
eConf C0303241 (2003) MOLT002 | arXiv:hep-ex/0305095 | 2003-05